Microplastic detection using spectral anomaly detection in oceans

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Introduction

- Microplastics (1mm–5mm) are one of the most widespread ocean pollutants
- Traditional monitoring is manual, expensive, and geographically limited
- Satellite remote sensing offers global, continuous coverage — but comes with a fundamental challenge

The challenge:

A single Sentinel-2 satellite pixel covers 10m × 10m of ocean surface. Microplastic particles are 1–5mm. You cannot see them.

Therefore: Detection must shift from visual pattern recognition to chemical spectral analysis — detecting the polymer fingerprint buried inside a pixel's aggregate reflectance signal.

Literature review

Lab methods identify plastics via polymer chemical signatures — accurate but unscalable to global ocean monitoring

Satellite hydrodynamic data confirms MP hotspots correlate with ocean current patterns — validating remote sensing as a detection medium

Surface satellite detection captures only the tip of the problem.

Existing satellite approaches have no learned spectral model for sub-pixel marine MP detection.

Dataset Characteristics

```
marida_dataset/
├── splits/
│   ├── train_X.txt  →  694 patches
│   ├── val_X.txt    →  328 patches
│   └── test_X.txt   →  359 patches
├── shapefiles/       (unused)
└── patches/
    └── S2_<scene>/
        ├── <patch>.tif      ← 11-band spectral data
        ├── <patch>_cl.tif   ← pixel-level class labels
        └── <patch>_conf.tif ← annotator confidence
```

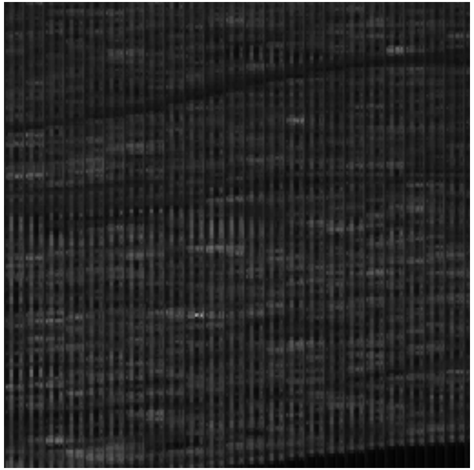
Dataset Characteristics

Each patch: 256 × 256 pixels, 11 spectral bands

15 class labels: Marine Debris, Marine Water, Turbid Water, Ships, Foam, Sargassum, etc.

Annotator confidence per pixel: 1=high, 2=medium, 3=low

Excluded conf=3 pixels from all training and evaluation



Model Definition

Feature Engineering — From 11 Bands to 14 Features

Why not just use raw bands?
Raw reflectance bands are correlated and don't directly encode the physics of plastic detection. We derive three indices that compress domain knowledge:

FAI — Floating Algae Index = $B8 - [B4 + (B11 - B4) \times (842 - 665) / (1610 - 665)]$

Measures NIR elevation above a Red-SWIR1 baseline. Water: FAI < 0. Floating material: FAI > 0.

PI — Plastic Index = $B8A / (B8A + B4 + \varepsilon)$

Polymers reflect strongly in NarrowNIR (~865nm), absorb in Red (~665nm). This ratio amplifies that contrast directly.

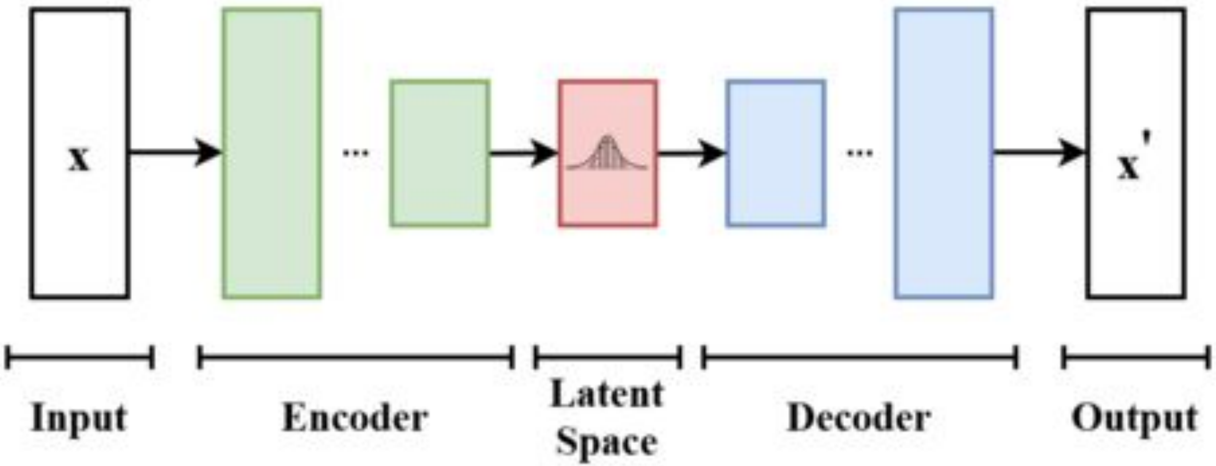
NDVI — Normalized Difference Vegetation Index = $(B8 - B4) / (B8 + B4 + \varepsilon)$

Separates vegetation (Sargassum) from water and plastic — important for reducing false positives.

Result: Each pixel is a 14-dimensional vector: 11 raw bands + FAI + PI + NDVI

Model Definition

VAE Architecture



Model Definition

```
class SpectralVAE(nn.Module):
    def encode(self, x):
        h      = self.encoder(x)      # 14 → 64 → 32
        mu     = self.fc_mu(h)        # 32 → 8
        logvar = self.fc_logvar(h)    # 32 → 8
        return mu, logvar
    def reparameterize(self, mu, logvar):
        if self.training:
            std = torch.exp(0.5 * logvar)
            eps = torch.randn_like(std)
            return mu + eps * std
        return mu
```

```
def forward(self, x):
    mu, logvar = self.encode(x)
    z          = self.reparameterize(mu, logvar)
    recon      = self.decode(z)    # 8 → 32 → 64 → 14
    return recon, mu, logvar
```

Model Definition

```
def elbo_loss(recon, x, mu, logvar, beta=0.5):
    recon_loss = F.mse_loss(recon, x, reduction='mean')
    kl_loss    = -0.5 * torch.mean(
                1 + logvar - mu.pow(2) - logvar.exp())
    return recon_loss + beta * kl_loss

scores = F.mse_loss(recon, x_norm,
                reduction='none').mean(dim=1)
#         higher score = more anomalous
```

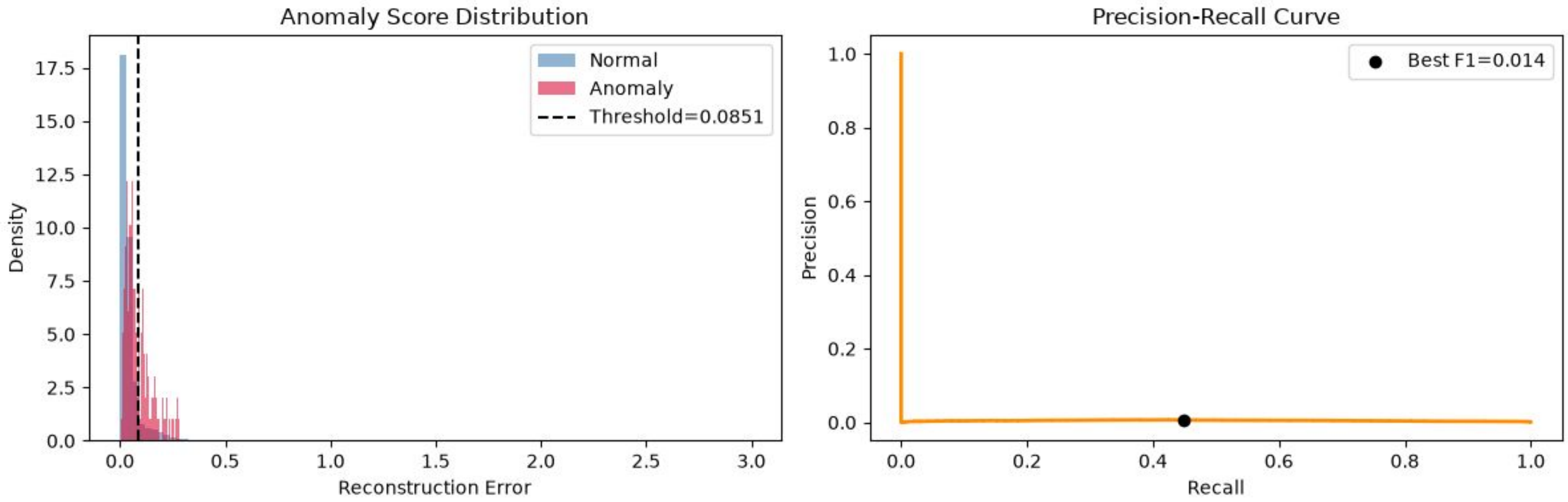
Results

Loss Curve summary:

Epoch	Train Loss	Val Loss	Val Recon	Val KL
1	0.445	0.196	0.084	0.223
10	0.188	0.171	0.046	0.249
25	0.184	0.169	0.043	0.252
50	0.182	0.170	0.041	0.258

Observations:
Loss converged cleanly — no divergence or instability
Val loss stable from epoch 5 onward — no overfitting
Reconstruction error steadily decreasing (0.084 → 0.041) — model improving
KL divergence gently rising (0.223 → 0.258) — healthy regularization, no posterior collapse

Results


Anomaly Score Distribution
Normal
Anomaly
Threshold=0.0851
Density
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Reconstruction Error
Precision-Recall Curve
Best F1=0.014
Precision
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
Recall

Results

ROC-AUC score in validation: 0.841

ROC-AUC score in test: 0.808

If we randomly pick one debris pixel and one normal pixel, the model assigns a higher anomaly score to the debris pixel **84.1% of the time** during training.

```
Val ROC-AUC  : 0.8412
Val Avg Prec : 0.0165
Test ROC-AUC  : 0.8081
Test Avg Prec : 0.0047
Best threshold      : 0.08511
Best F1             : 0.0137
Precision @ best F1 : 0.0070
Recall    @ best F1 : 0.4479
Saved evaluation.png
```

Challenges

99.8% of training pixels are Class 0 "Unknown" — unannotated, clean water. The VAE learned a broad manifold over these unlabeled pixels. Debris pixels, deviating only subtly, were not enough, which is why VAE Couldn’t learn enough features for the MP..

Attempted fix:
Tried restricting training to confirmed classes only (7, 8, 10, 11, 15). This caused a train/val distribution mismatch — the model had never seen the heterogeneous val/test pixels, so normal non-water pixels scored as anomalies. Train loss: 0.365, Val loss: 0.915 — large generalization gap.
ROC-AUC dropped to 0.698.

Conclusion

Identified problem:

99.8% of training pixels are Class 0 "Unknown"

What we achieved:

- Built a sub-pixel anomaly detection pipeline for marine microplastics
- Sentinel-2 spectral data → physics-informed features → VAE → anomaly score per pixel
- Val ROC-AUC of 0.841 with **zero labeled anomaly examples during training**

Future Work

1. Water-anchored normalization *(immediate)*
Recompute input normalization stats from confirmed water pixels to tighten the normal manifold without causing distribution mismatch.

2. Spectral band weighting in loss *(near-term)*
Upweight B8, B8A, B11, B12, FAI, PI in the reconstruction loss — forces the model to focus capacity on polymer-discriminative features.

3. Multi-scene generalization *(long-term)*
Test on held-out geographic regions to assess whether the model generalizes across different ocean conditions, seasons, and atmospheric states.

Thank You !